

Zooplankton collection (DP1.20219.001)

Measurement

Taxonomic identification and counts per taxon and per size class of zooplankton from the water column of lakes.

Collection methodology

Zooplankton are collected at lake sites three times annually (roughly spring, summer and fall). Samples are collected at the inlet, outlet and buoy (deepest) locations using a vertical tow net or a Schindler-Patalas sampler, depending on water depth. Typically, multiple samples (up to 3) are collected per sample location and composited into a single sample and total water volume is recorded. Composite samples are preserved with ethanol in the field and sent to an external facility for taxonomic identification.

For information about disturbances, land management activities, and other incidents that may impact data at NEON sites, see the [Site management and event reporting \(DP1.10111.001\)](#) data product.

Data package contents

zoo_taxonomyProcessed: Aquatic zooplankton identifications by expert taxonomists - desynonymized

zoo_taxonomyRaw: Aquatic zooplankton identifications by expert taxonomists - raw

zoo_fieldData: Aquatic zooplankton field data

zoo_perSample: Zooplankton subsampling data and QC metrics per sample

zoo_perVial: Aquatic zooplankton identified archive data

zoo_identificationHistory: Aquatic zooplankton identification history for records where identifications have changed

variables: Description and units for each column of data in data tables

readme: Data product description, issue log, and other metadata about the data product

validation: Description of data validation applied at the points of collection and ingest

Data quality

Ten percent of all samples are quality checked for taxonomic difference and difference in enumeration between two taxonomists at the external facility. These records are indicated by the fields qcChecked, qcEnumerationDifference, and qcTaxonomicDifference, and data are corrected if necessary prior to return to NEON.

Please note that quality checks are comprehensive but not exhaustive; therefore, unknown data quality issues may exist. Users are advised to evaluate quality of the data as relevant to the scientific research question being addressed, perform data review and post-processing prior to analysis, and use the data quality information and issue logs included in download packages to aid interpretation.

Standard calculations

For wrapper functions to download data from the API, and functions to merge tabular data files across sites and months, NEON provides the `neonUtilities` package in R and the `neonutilities` package in Python. See the [Download and Explore NEON Data](#) tutorial for introductory instructions in both programming languages.

For quantitative results, follow these steps prior to analyzing data:

1. Join expert taxonomy and field data tables

```
zoo_join <- neonOS::joinTableNEON(zoo_taxonomyProcessed, zoo_fieldData)
```

2. Calculate zooplankton abundance per liter

```
zoo_join$abunPerL <- zoo_join$adjCountPerBottle / zoo_join$stowsTrapsVolume
```

Table joining

Table 1	Table 2	Join by field(s)
zoo_fieldData	zoo_perSample	sampleID
zoo_perSample	zoo_taxonomyRaw	sampleID
zoo_perSample	zoo_taxonomyProcessed	sampleID
zoo_fieldData	zoo_taxonomyProcessed	sampleID
zoo_fieldData	zoo_taxonomyRaw	sampleID
zoo_perSample	zoo_perVial	sampleID
zoo_perVial	zoo_taxonomyProcessed	Join not recommended. Data resolution does not match other tables.

Table 1	Table 2	Join by field(s)
zoo_perVial	zoo_taxonomyRaw	Join not recommended. Data resolution does not match other tables.
zoo_taxonomyProcessed	zoo_taxonomyRaw	Join not recommended. If interested in processed NEON vs. raw lab taxonomy, may join on sample ID + scientificName.

Documentation

-  [Standard Operating Procedure Laboratory Methods: NEON-Zooplankton Version 1.1](#)
LIMNOPRO_Zooplankton_ID_Rev1.1 | 625.1 KiB | PDF
-  [NEON Aquatic Sampling Strategy](#)
NEON.DOC.001152vD | 846.5 KiB | PDF
-  [AOS Protocol and Procedure: ZOO – Zooplankton Sampling in Lakes](#)
NEON.DOC.001194vL | 1.1 MiB | PDF
-  [NEON Algorithm Theoretical Basis Document \(ATBD\): OS Data Quality Control](#)
NEON.DOC.005424vA | 651.3 KiB | PDF
-  [NEON User Guide to Aquatic Zooplankton Collection \(NEON.DP1.20219\)](#)
NEON_zooplankton_userGuide_vG | 1.4 MiB | PDF

For more information on data product documentation, see:
<https://data.neonscience.org/data-products/DP1.20219.001>

Citation

To cite data from value: "Zooplankton collection" (DP1.20219.001), see citation here:
<https://data.neonscience.org/data-products/DP1.20219.001>
For general guidance in citing NEON data and documentation, see the citation guidelines page:
<https://www.neonscience.org/data-samples/guidelines-policies/citing>

Contact Us

NEON welcomes discussion with data users! Reach out with any questions or concerns about NEON data:
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